

2030 DRAM Sıkışması: Yapay Zeka

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User prompt: # The 2030 DRAM Squeeze: 157.5 Exabytes of Demand Meets Only 128.8 Exabytes of Supply ## Executive Summary The global memory market faces an unprecedented supply-demand imbalance by 2030, with projected demand reaching 157.5 exabytes (EB) while base-case supply is expected to reach only 128.8 EB. This 28.7 EB shortfall, representing approximately 24 percent of general DRAM demand, is driven primarily by the explosive growth of artificial intelligence workloads, particularly those requiring high-bandwidth memory (HBM) and the emergence of agentic AI systems that demand significantly more CPU and DRAM resources. This structural imbalance is not a temporary market fluctuation but a fundamental shift in memory architecture requirements, with HBM demand fully embedded within the 157.5 EB total while general DRAM remains the primary bottleneck. Industry leaders including Samsung, SK Hynix, and Micron are accelerating fabrication plant expansions, yet meaningful capacity relief is unlikely before 2028, with the shortage expected to persist through the end of the decade. --- ## Introduction The semiconductor memory market is experiencing a once-in-a-generation transformation. Traditional computing paradigms, which relied heavily on GPU acceleration for AI tasks, are evolving toward agentic AI systems that require balanced CPU-GPU architectures. This evolution, combined with the insatiable demand for high-bandwidth memory in AI training and inference, has created a perfect storm in the DRAM market. According to industry projections, 2030 will see annual DRAM demand reach 157.5 EB, outstripping supply by nearly 30 EB. This shortfall is not merely a numerical discrepancy; it represents a fundamental mismatch between the memory requirements of next-generation AI systems and the current manufacturing capabilities of the world's leading memory producers. --- ## Market Demand Analysis #### Total Demand Projection: 157.5 EB The 2030 DRAM demand landscape is dominated by five key categories, each representing distinct use cases and technological requirements: ##### 1. Agentic CPUs: 75.0 EB (47.6 percent) Agentic AI represents the most significant demand driver, accounting for nearly half of all projected DRAM consumption. Unlike traditional AI systems that rely primarily on GPU acceleration, agentic AI requires substantial CPU resources to handle complex, autonomous tasks. These systems perform functions such as code inspection, supply chain coordination, and financial research synthesis, fundamentally altering the computational architecture of data centers. The architectural shift is substantial: CPU-to-GPU ratios are moving from historical levels of 1:4 toward 1:1 or even 1:2 configurations. This represents a 2-4x increase in CPU demand, with corresponding memory requirements. Industry reports indicate that CPUs running agentic AI workloads may require up to 400 GB of memory per unit, four times the capacity of current systems. ##### 2. General Cloud / IaaS CPUs: 25.0 EB (15.9 percent) Traditional cloud computing and Infrastructure-as-a-Service (IaaS) platforms continue to drive significant DRAM demand. While not growing at the same rate as AI-specific workloads, these established use cases maintain substantial memory requirements for enterprise applications, virtualization, and general-purpose computing. The cloud segment benefits from the overall growth in data center deployment but faces competition for memory resources as manufacturers prioritize more profitable HBM production over commodity DRAM. ##### 3. HBM5: 22.5 EB (14.3 percent) High-Bandwidth Memory 5 (HBM5) represents the cutting edge of memory technology, designed specifically for AI accelerators and next-generation GPU systems. HBM5 offers dramatically higher bandwidth and lower power consumption compared to traditional DRAM, making it essential for AI training workloads. The transition to HBM5, expected in the 2028-2029 timeframe, will further concentrate wafer capacity in advanced processes, potentially exacerbating supply constraints for commodity DRAM as manufacturers reallocate resources toward these premium products. ##### 4. Consumer DRAM: 20.0 EB (12.7 percent) Consumer applications, including PCs, smartphones, and gaming systems, continue to represent a significant portion of DRAM demand. However, this segment is increasingly squeezed as memory manufacturers prioritize AI and data center

customers who offer higher margins and long-term supply agreements. The consumer DRAM market faces particular pressure as HBM production consumes disproportionate wafer capacity. Industry analysis reveals that 1GB of HBM requires approximately 4x the manufacturing capacity of standard DRAM, creating a structural disadvantage for consumer memory production. ##### 5. HBM4E: 15.0 EB (9.5 percent) HBM4E, an enhanced version of HBM4, represents an intermediate step in the memory technology evolution. Expected to enter production in late 2027, HBM4E provides improved performance over HBM3E while serving as a bridge to the more advanced HBM5 standard. Like other HBM variants, HBM4E production consumes significant advanced manufacturing capacity, contributing to the overall supply constraints while meeting the growing demands of AI workloads. --- ## Supply Analysis

Base-Case Supply: 128.8 EB The projected supply of 128.8 EB for 2030 is composed of two primary categories: ##### General DRAM: 91.3 EB General DRAM, which includes standard DDR4, DDR5, and LPDDR variants, constitutes the majority of supply at 91.3 EB. However, this falls significantly short of the projected general DRAM demand of 120.0 EB, creating a 28.7 EB shortfall. This discrepancy represents approximately 24 percent of general DRAM demand going unmet, making general DRAM the primary bottleneck in the 2030 memory market. The shortfall is particularly acute because general DRAM serves as the foundation for most computing applications, from consumer devices to enterprise servers. ##### HBM: 37.5 EB High-Bandwidth Memory production is projected to reach 37.5 EB by 2030, representing the entire HBM component of the demand equation. Unlike general DRAM, HBM demand is fully embedded within the 157.5 EB total, meaning that all HBM production is accounted for within the overall demand figure. HBM production faces unique challenges, including: - Lower Manufacturing Yields: 12-layer HBM3e stacks currently achieve yields of only 50-60 percent, significantly lower than standard DRAM yields of greater than 90 percent - Advanced Packaging Requirements: HBM requires through-silicon via (TSV) technology and sophisticated thermal management, limiting production to specialized facilities - Resource Intensity: As previously noted, HBM consumes 4x the wafer capacity of standard DRAM per gigabyte produced --- ## Manufacturing Capacity Expansion

Current Expansion Efforts The world's three major DRAM manufacturers—Samsung, SK Hynix, and Micron—are engaged in an unprecedented capacity expansion race to address the growing demand: ##### SK Hynix SK Hynix has announced the most aggressive expansion plans, aiming to nearly double its DRAM wafer production capacity: - Current Capacity: Approximately 550,000 wafers per month (including 200,000 from Wuxi, China) - 2030 Target: 1 million wafers per month - Key Projects: - Yongin semiconductor cluster: First fab divided into six cleanrooms, with equipment installation beginning February 2027 - Each cleanroom adds 60,000 monthly wafer starts, with full 360,000-wafer capacity expected by first half of 2030 - M15X fab in Cheongju: Scheduled to begin operations in second half of 2026 with initial capacity of 40,000 wafers/month, reaching 80,000 in 2027 - Investment: Additional USD 15 billion approved for first Yongin fab by end of 2030 ##### Samsung Samsung has accelerated its expansion timeline significantly: - Yongin National Industrial Complex: First of six planned fabs moved up from 2030-2031 to October 2029 - Investment: USD 400 trillion won (approximately USD 266.9 billion) for two new memory fabs in South Korea's Honam region - Collaborations: Signed letter of intent with SK Hynix and OpenAI for eventual supply of 900,000 DRAM wafers per month to meet OpenAI's Stargate project demands ##### Micron Micron is pursuing a multi-regional diversification strategy: - Idaho Fab: First wafer output expected in mid-2027 - Second Idaho Fab: Targeted for end of 2028 - New York Fab: Supply expected in 2030 and beyond - Singapore: HBM advanced packaging facility expected to contribute meaningfully in 2027 - Investment: USD 250 billion investment in US facilities by 2035, with 40 percent of DRAM production targeted for the United States

Global Capacity Projections Industry analysis suggests that global DRAM capacity is expected to grow at an average annual rate of 4.8 percent through 2030, reaching approximately 2 million 12-inch wafers per month. However, this growth rate may prove insufficient to meet the projected demand, particularly given the resource-intensive nature of HBM production. --- ## Key Market Dynamics

The Agentic AI Revolution Agentic AI represents a fundamental shift in computational requirements. Unlike traditional AI systems that perform static inference tasks, agentic systems are dynamic, autonomous, and capable of complex decision-making.

This evolution has several implications for memory demand: 1. CPU Resurgence: Agentic workloads require significantly more CPU resources, with CPU-to-GPU ratios shifting from 1:4 to 1:1 or 1:2 2. Memory Intensity: Individual CPUs may require 400 GB of memory, four times current levels 3. Architectural Complexity: Agentic systems need a complete memory hierarchy, from latency-optimized SRAM through HBM, GDDR, DDR, CXL, and persistent context storage

Micron Technology identifies agentic AI as driving a "resurgence in CPU and DRAM demand," with memory now on the critical path for AI infrastructure development. ### HBM Market Growth The HBM market is experiencing exponential growth, driven by AI workloads: - SK Hynix Forecast: 30 percent annual growth until 2030 - Micron Projection: HBM total addressable market reaching USD 100 billion by 2028 (originally forecast for 2030) - Market Composition: AI applications consume the vast majority of HBM supply, with automotive emerging as the fastest-growing segment - Technological Evolution: Rapid progression through HBM generations, with HBM4E expected in late 2027 and HBM5 in 2028-2029

Supply Chain Constraints Several structural constraints limit the industry's ability to respond to growing demand: 1. Manufacturing Complexity: HBM production requires advanced packaging, TSV technology, and intricate thermal management 2. Equipment Limitations: Extreme ultraviolet lithography equipment and specialist gases are in limited supply 3. Geopolitical Factors: US export controls limit the ability of Samsung and SK Hynix to expand Chinese facilities 4. Yield Challenges: HBM manufacturing yields remain significantly lower than standard DRAM 5. Capacity Allocation: Manufacturers are reallocating wafer starts from commodity DRAM to HBM, tightening availability for other sectors --- ## Price and Availability Impact The supply-demand imbalance has already resulted in significant market effects: - Price Increases: DRAM prices rose 60 percent in 2025, with an additional 30-40 percent increase projected for 2026 - Allocation Systems: Memory suppliers are implementing longer supply agreements and tighter allocation mechanisms - Market Segmentation: The gap between players with locked supply and those without is widening - Consumer Impact: PC, gaming, and consumer electronics markets face particular pressure as manufacturers prioritize AI data center customers Industry analysts note that the current build-out, if completed, could eventually meet pent-up demand and restore price equilibrium. However, this equilibrium is unlikely to be achieved before 2028-2029, with the highest-end HBM and advanced packaging markets potentially remaining allocation-driven through 2030. --- ## Industry Perspective and Timeline ### Short-Term (2026-2027) The immediate future will see continued tightening of memory supply: - SK Group Chairman Chey Tae-won has stated that "until 2030 there's still some shortage" - Micron CEO Sanjay Mehrotra warned that meaningful large-scale new capacity is unlikely before 2028 - Current capacity meets only 50-66 percent of core customer demand ### Medium-Term (2028-2029) Gradual relief is expected as new fabrication facilities come online: - First Yongin fab reaches full capacity by first half of 2030 - Micron's Idaho and Singapore facilities begin contributing - Samsung's accelerated fab construction provides additional supply - More comfortable supply environment for DDR5, LPDDR server memory, and HBM ### Long-Term (2030 and Beyond) The industry faces a critical junction: - If agentic AI adoption accelerates faster than capacity expansion, 2028 may only shift the shortage from severe to manageable - Large fab projects and supplier capacity expansions become more meaningful by 2029-2030 - Micron expects New York supply to contribute significantly post-2030 - The structural nature of the shortage suggests it will not fully resolve before the end of the decade --- ## Technological Context ### Memory Hierarchy for Agentic AI Modern agentic AI systems require a comprehensive memory hierarchy: 1. SRAM: Latency-optimized for fastest access 2. HBM: High-bandwidth, low-latency memory for AI accelerators 3. GDDR: Graphics-focused memory for high-performance computing 4. DDR: Standard DRAM for general computing 5. CXL: Compute Express Link for memory expansion 6. Persistent Storage: For context and long-term data retention Each layer serves specific purposes, with HBM and DDR representing the most critical components for AI workloads. ### HBM Generational Progression The rapid evolution of HBM technology demonstrates the industry's response to AI demands: - HBM3: Current generation, offering significant bandwidth improvements - HBM3E: Enhanced version with higher capacity and bandwidth - HBM4: Next-generation standard, expected in production by 2027 - HBM4E: Enhanced HBM4 variant, expected late 2027 - HBM5:

Advanced standard, expected 2028-2029 Each generational transition concentrates more wafer capacity in lower-bit-yield processes, potentially creating additional supply shocks rather than relief. --- ## Market Share and Competitive Landscape ### Current Market Position (Q1 2026) - Samsung: 38 percent DRAM market share - SK Hynix: 29 percent DRAM market share - Micron: 22.5 percent DRAM market share (11 percent noted in some reports) ### HBM Market Position - SK Hynix: Leading HBM supplier, recently overtook Samsung for the number two spot with 23 percent share - Samsung: Strong competitor in HBM market - Micron: Gaining significant HBM market share, expected to be a major beneficiary of US CHIPS Act funding ### Investment Scale The three major manufacturers are engaged in a "capacity expansion war": - Samsung and SK Hynix: Joint investment of USD 518 billion to build four new memory fabs and an HBM hub in Korea - SK Hynix: USD 26.7 billion raised from Nasdaq listing for domestic and international expansion - Micron: USD 250 billion investment in US facilities by 2035 --- ## Strategic Implications ### For Memory Manufacturers 1. Capacity Prioritization: Continue shifting wafer starts toward HBM and advanced DDR5 products 2. Technology Investment: Accelerate development of next-generation memory technologies 3. Geographic Diversification: Expand manufacturing presence across multiple regions to mitigate geopolitical risks 4. Partnership Strategy: Develop long-term supply agreements with key AI customers ### For AI Companies 1. Supply Security: Secure long-term memory supply agreements to ensure access to critical components 2. Architectural Optimization: Design systems to optimize memory usage and reduce dependency on constrained resources 3. Diversification: Explore alternative memory technologies and suppliers to mitigate supply chain risks 4. Investment Planning: Account for continued memory price premiums in infrastructure budgeting ### For End Users 1. Expect Higher Costs: Prepare for continued memory price increases through 2030 2. Plan for Delays: Anticipate longer lead times for memory-intensive systems 3. Consider Alternatives: Evaluate alternative architectures and technologies that may be less affected by supply constraints 4. Strategic Stockpiling: For critical applications, consider strategic inventory management --- ## Conclusion The 2030 DRAM Squeeze represents a fundamental structural shift in the semiconductor memory market. Driven by the explosive growth of agentic AI and the resource-intensive nature of HBM production, the industry faces an unprecedented supply-demand imbalance that will persist through the end of the decade. With projected demand of 157.5 EB outstripping supply of 128.8 EB, the market faces a 28.7 EB shortfall that cannot be resolved through current expansion plans alone. The situation is exacerbated by the technical challenges of HBM manufacturing, the architectural evolution toward agentic AI systems, and the geopolitical constraints on capacity expansion. While the industry's major players—Samsung, SK Hynix, and Micron—are investing hundreds of billions of dollars in new fabrication facilities, meaningful relief is unlikely before 2028. The structural nature of this shortage, combined with the continued evolution of AI workloads, suggests that memory will remain on the critical path for technological advancement well into the next decade. The 2030 DRAM Squeeze is not merely a market fluctuation; it is a fundamental redefinition of memory requirements in the age of artificial intelligence. Successfully navigating this challenge will require coordinated effort across the entire technology ecosystem, from memory manufacturers to AI developers to end users. --- ## Sources and References 1. IDC Memory Market Outlook (2026) - Worldwide memory demand and supply forecasts through 2030 2. SK Hynix Corporate Announcements (2025-2026) - Capacity expansion plans and HBM growth forecasts 3. Micron Technology Investor Presentations (2025-2026) - HBM market projections and capacity timeline 4. Samsung Electronics Corporate Reports (2025-2026) - Fab construction acceleration and investment plans 5. TrendForce Industry Analysis (2025-2026) - DRAM market share and supply-demand dynamics 6. Commercial Times Industry Reports (2025-2026) - AI memory consumption projections 7. SemiWiki Industry Analysis (2026) - Global DRAM capacity growth projections 8. Forbes Technology Coverage (2026) - Memory capacity production expansion analysis 9. The Elec Industry Reports (2026) - SK Hynix capacity roadmap and expansion details 10. IntuitionLabs Market Analysis (2025) - RAM shortage and AI demand impact assessment *Note: All figures referenced in this article are modeled estimates for 2030. Demand categories are mutually exclusive and sum to the total 157.5 EB projection.* yazılanları araştır ve bir makale üret

Response: 2030 DRAM Sıkışması: Yapay Zeka Çağında Hafıza Kısılacı

Yönetici Özeti

Küresel teknoloji pazarı, 2030 yılına doğru ilerlerken eşi benzeri görülmemiş bir arz-talep dengesizliğiyle karşı karşıya kalmaya hazırlanıyor. Sektörel projeksiyonlar, 2030 yılında küresel DRAM talebinin **157.5 Ekabayt (EB)** seviyesine ulaşacağını öngörürken, mevcut ve planlanan bazı senaryo arzının ancak **128.8 EB** düzeyinde kalacağını gösteriyor. Ortaya çıkan **28.7 EB'lık devasa açık**, genel DRAM talebinin yaklaşık yüzde 24'ünün karşılanamaması anlamına geliyor.

Bu yapısal krizin arkasında, yapay zeka modellerinin yalnızca eğitime (training) aşamasından, otonom kararlar alabilen **"Ajan Yapay Zeka" (Agentic AI)** ve çıkarım (inference) odaklı sistemlere geçiş yapması yatıyor. Bu yeni mimari, hem GPU hem de CPU tarafında muazzam bir hafıza bant genişliği ve kapasitesi talep ediyor. Sektör devleri Samsung, SK Hynix ve Micron fabrika genişletme çalışmalarına hız verse de 2028 yılından önce pazarda anlamlı bir rahatlama beklenmiyor.

Pazar Talep Analizi: 157.5 EB Nereye Harcanacak?

2030 yılındaki DRAM tüketim senaryoları incelendiğinde, talebin neredeyse yarısının tek bir dinamik tarafından domine edildiği görülüyor: Ajan Yapay Zekaya sahip işlemciler.

2030 DRAM Talep Dağılımı

Kategori	Talep Miktarı (EB)	Pazar Payı (%)
Ajan İşlemciler (Agentic CPUs)	75.0	47.6%
Genel Bulut / IaaS İşlemcileri	25.0	15.9%
HBM5 (High-Bandwidth Memory 5)	22.5	14.3%
Tüketici Elektronik DRAM	20.0	12.7%
HBM4E	15.0	9.5%

Kategori	Talep Miktarı (EB)	Pazar Payı (%)
Toplam	157.5	100%

Temel Talep İtici Güçleri

- Ajan Yapay Zeka Devrimi (Agentic CPUs):** Geleneksel yapay zekanın aksine, otonom kod denetimi, finansal analiz veya tedarik زنجيري yönetimi gibi karmaşık ve sürekli tekrarlayan döngüleri yöneten ajan sistemler, CPU-GPU oranlarını geçmişteki 1:4 seviyelerinden 1:1 veya 1:2 seviyelerine çekiyor. Sektör raporlarına göre, ajan yapay zeka iş yüklerini çalıştıran tek bir CPU'nun hafıza ihtiyacı **400 GB** seviyesine (mevcut sistemlerin 4 katı) çıkabilir.
- HBM (Yüksek Bant Genişlikli Hafıza) Geçişi:** HBM4E ve özellikle 2028-2029 döneminde devreye girmesi beklenen HBM5 teknolojileri, AI hızlandırıcılarının kalbini oluşturuyor. Ancak bu premium ürünlerin üretimi, standart DRAM hatlarını adeta "yutuyor".
- Tüketici Eğilimleri:** Akıllı telefonlar, bilgisayarlar ve oyun konsolları hâlâ pazarın yüzde 12.7'sini oluştursa da üreticilerin yüksek marjlı yapay zeka müşterilerine öncelik vermesi nedeniyle bu segment ciddi şekilde gölgede kalıyor.

Arz Analizi ve Yapısal Tıkanıklık

2030 yılı için öngörülen **128.8 EB'lık** toplam arz iki ana kategoriye ayrılıyor:

- Genel DRAM (DDR4, DDR5, LPDDR):** 91.3 EB
- HBM Üretimi:** 37.5 EB

Krizin Esas Kaynağı: Genel DRAM pazarındaki talep 120.0 EB seviyesindeyken, üretilebilecek miktar 91.3 EB'da kalıyor. Bu durum, genel amaçlı sunuculardan akıllı telefonlara kadar tüm ekosistemi vuracak olan **28.7 EB'lık genel DRAM darboğazını** doğuruyor.

HBM Üretiminin Standart Belleği Baskılaması (Cannibalization)

HBM üretimi, standart DRAM'e kıyasla oldukça agresif kaynak tüketiyor:

- Kapasite Tüketimi:** 1 GB HBM üretmek, standart bir DRAM'e göre yaklaşık **4 kat daha fazla** silikon gofret (wafer) kapasitesi harcıyor.

2. **Düşük Verimlilik (Yield Rates):** Mevcut 12 katmanlı HBM3e üretiminde verimlilik oranları yüzde 50-60 civarında seyrederken, standart DRAM'lerde bu oran yüzde 90'ın üzerinde.
3. **Gelişmiş Paketleme İhtiyacı:** Silikon içi dikey hatlar (TSV) ve karmaşık termal yönetim gereksinimleri, üretimi yalnızca belirli özel tesislere hapsediyor. Üreticiler kâr marjı yüksek olan HBM segmentine öncelik verdikçe, standart DDR5 ürünlerinin kapasitesi daralıyor.

Üretim Kapasitesi Savaşları: Üç Büyüklerin Hamleleri

Pazardaki bu devasa açığı kapatmak adına Samsung, SK Hynix ve Micron milyarlarca dolarlık yatırım yarışına girmiş durumda.

- **SK Hynix:** Aylık 550 bin olan gofret (wafer) üretim kapasitesini 2030'a kadar **1 milyona** çıkarmayı hedefliyor. Yongin yarı iletken kümelenmesindeki ilk fabrikanın kurulumuna Şubat 2027'de başlanacak ve tam kapasiteye 2030'un ilk yarısında ulaşılacak. Ayrıca Cheongju'daki M15X fabrikası 2026'nın ikinci yarısında devreye giriyor.
- **Samsung:** Yongin Ulusal Endüstriyel Kompleksi'ndeki altı planlı fabrikanın ilkinin açılış tarihini Ekim 2029'a çekti. OpenAI'nın Stargate projesi gibi devasa altyapı taleplerini karşılamak adına SK Hynix ve OpenAI ile niyet mektupları imzalayan şirket, iki yeni fabrika için yaklaşık **266.9 milyar dolar** bütçe ayırdı.
- **Micron:** Coğrafi çeşitlilik stratejisi izleyen şirket, Idaho'daki ilk fabrikasından 2027 ortasında ürün almayı bekliyor. New York fabrikasının ise 2030 ve sonrasında devreye girmesi planlanıyor. Micron, 2035 yılına kadar ABD tesislerine **250 milyar dolar** yatırım yapmayı taahhüt etti.

Pazar Dinamikleri ve Stratejik Çıkarımlar

Fiyatlama ve Erişilebilirlik

Hafıza pazarında yaşanan bu yapısal dönüşüm, geçici bir dalgalanma değil. 2025 yılında DRAM fiyatlarında görülen yüzde 60'lık artışın, 2026'da yüzde 30-40 seviyesinde ek zamlarla devam etmesi bekleniyor. Üreticiler spot piyasalar yerine uzun vadeli tedarik sözleşmelerine (LTA) ve büyük bulut sağlayıcılarına (CSP) öncelik veriyor. Bu durum tüketici elektroniği ve oyun bilgisayarları pazarlarını ciddi bir arz kıskacına sokuyor.

Sektör Paydaşları İçin Eylem Planı

- **Yapay Zeka Şirketleri İçin:** Donanım altyapısını garanti altına almak adına şimdiden 3 ila 5 yıllık uzun vadeli hafıza tahsis sözleşmeleri yapılmalı; KV cache optimizasyonları ve verimli bellek hiyerarşileri (CXL, SRAM entegrasyonları) yazılımsal olarak geliştirilmelidir.

- **Son Kullanıcılar ve Kurumsal Alıcılar İçin:** Sunucu ve veri merkezi bütçelerinde bellek maliyet kalemlerinin ağırlığı artırılmalı, kritik donanım yenileme projelerinde teslimat sürelerinin (lead time) uzayacağı hesaba katılmalıdır.

Sonuç

2030 DRAM Sıkışması, yarı iletken dünyasının sadece geçici bir üretim döngüsü sorunu değil; yapay zekanın mimari evriminin (özellikle çıkarım ve ajan iş yüklerinin) sektöre ödettiği yapısal bir faturadır. Büyük üreticilerin yüz milyarlarca dolarlık yatırımları pazara ancak 2028-2029 döneminden sonra anlamlı bir rahatlama getirebilecektir. O zamana kadar bellek, yapay zeka ve genel bilgi işlem ekosisteminin en kritik ve pahalı darboğazı olmaya devam edecektir.
